

The Second Fundamental Theorem of Calculus

Theorem: Let f be a continuous function on the closed interval $[a, b]$.
If $a \leq c \leq b$, then for x in $[a, b]$,

$$D_x \int_c^x f(t) dt = f(x)$$

1) Let $F(x) = \int_0^x (t^2 + 1) \cdot dt$

Hint: $\int (t^2 + 1) \cdot dt = \frac{t^3}{3} + t$

$$F(x) = \int_0^x (t^2 + 1) \cdot dt = \left[\frac{t^3}{3} + t \right]_0^x = \left[\frac{x^3}{3} + x \right] - \left[\frac{0^3}{3} + 0 \right] = \frac{x^3}{3} + x$$

a) $F(1) = \int_0^1 (t^2 + 1) \cdot dt = \underline{\hspace{2cm}} ?$

b) $F(2) = \int_0^2 (t^2 + 1) \cdot dt = \underline{\hspace{2cm}} ?$

c) $F(3) = \int_0^3 (t^2 + 1) \cdot dt = \underline{\hspace{2cm}} ?$

d) $F'(x) = D_x \left[\int_0^x (t^2 + 1) \cdot dt \right] = D_x \left[\frac{x^3}{3} + x \right] = \underline{\hspace{2cm}} ?$

e) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^2 + 1) \cdot dt \right] = \underline{\hspace{2cm}} ?$

2) Let $F(x) = \int_0^x (t^3 + 5) \cdot dt$

Hint: $\int (t^3 + 5) \cdot dt = \frac{t^4}{4} + 5t$

$$F(x) = \int_0^x (t^3 + 5) \cdot dt = \left[\frac{t^4}{4} + 5t \right]_0^x = \left[\frac{x^4}{4} + 5x \right] - \left[\frac{0^4}{4} + 5 \cdot 0 \right] = \frac{x^4}{4} + 5x$$

a) $F(1) = \int_0^1 (t^3 + 5) \cdot dt = \underline{\hspace{2cm}} ?$

b) $F(2) = \int_0^2 (t^3 + 5) \cdot dt = \underline{\hspace{2cm}} ?$

c) $F(3) = \int_0^3 (t^3 + 5) \cdot dt = \underline{\hspace{2cm}} ?$

d) $F'(x) = D_x \left[\int_0^x (t^3 + 5) \cdot dt \right] = D_x \left[\frac{x^4}{4} + 5x \right] = \underline{\hspace{2cm}} ?$

e) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^3 + 5) \cdot dt \right] = \underline{\hspace{2cm}} ?$

3) Let $F(x) = \int_0^x (t^2 - 5t + 1) \cdot dt$

Hint: $\int (t^2 - 5t + 1) \cdot dt = \frac{t^3}{3} - \frac{5t^2}{2} + t$

$$F(x) = \int_0^x (t^2 - 5t + 1) \cdot dt = \left[\frac{t^3}{3} - \frac{5t^2}{2} + t \right]_0^x = \left[\frac{x^3}{3} - \frac{5x^2}{2} + x \right] - \left[\frac{0^3}{3} - \frac{5 \cdot 0^2}{2} + 0 \right] = \frac{x^3}{3} - \frac{5x^2}{2} + x$$

a) $F(1) = \int_0^1 (t^2 - 5t + 1) \cdot dt = \underline{\hspace{2cm}} ?$

b) $F(2) = \int_0^2 (t^2 - 5t + 1) \cdot dt = \underline{\hspace{2cm}} ?$

c) $F(3) = \int_0^3 (t^2 - 5t + 1) \cdot dt = \underline{\hspace{2cm}} ?$

d) $F'(x) = D_x \left[\int_0^x (t^2 - 5t + 1) \cdot dt \right] = D_x \left[\frac{x^3}{3} - \frac{5x^2}{2} + x \right] = \underline{\hspace{2cm}} ?$

e) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^2 - 5t + 1) \cdot dt \right] = \underline{\hspace{2cm}} ?$

4) Let $F(x) = \int_1^x (\sqrt{t} + 4) \cdot dt$

Hint: $\int (\sqrt{t} + 4) \cdot dt = \int (t^{1/2} + 4) \cdot dt = \frac{t^{3/2}}{3/2} + 4t = \frac{2}{3} \cdot t^{3/2} + 4t$

$$F(x) = \int_1^x (\sqrt{t} + 4) \cdot dt = \left[\frac{2}{3} \cdot t^{3/2} + 4t \right]_1^x = \left[\frac{2}{3} \cdot x^{3/2} + 4x \right] - \left[\frac{2}{3} \cdot (1)^{3/2} + 4(1) \right] = \frac{2}{3} \cdot x^{3/2} + 4x - \frac{14}{3}$$

a) $F(1) = \int_1^1 (\sqrt{t} + 4) \cdot dt = \underline{\hspace{2cm}} ?$

b) $F(2) = \int_1^2 (\sqrt{t} + 4) \cdot dt = \underline{\hspace{2cm}} ?$

c) $F(3) = \int_1^3 (\sqrt{t} + 4) \cdot dt = \underline{\hspace{2cm}} ?$

d) $F'(x) = D_x \left[\int_1^x (\sqrt{t} + 4) \cdot dt \right] = D_x \left[\frac{2}{3} \cdot x^{3/2} + 4x - \frac{14}{3} \right] = \underline{\hspace{2cm}} ?$

e) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_1^x (\sqrt{t} + 4) \cdot dt \right] = \underline{\hspace{2cm}} ?$

5) Let $F(x) = \int_1^x \frac{1}{t^2} \cdot dt$

Hint: $\int \frac{1}{t^2} \cdot dt = \int t^{-2} \cdot dt = \frac{t^{-1}}{-1} = -t^{-1} = \frac{-1}{t}$

$$F(x) = \int_1^x \frac{1}{t^2} \cdot dt = \frac{-1}{x} - \frac{-1}{1} = \frac{-1}{x} + 1$$

a) $F(1) = \int_1^1 \frac{1}{t^2} \cdot dt = \underline{\hspace{2cm}}$

b) $F(2) = \int_1^2 \frac{1}{t^2} \cdot dt = \underline{\hspace{2cm}}$

c) $F(3) = \int_1^3 \frac{1}{t^2} \cdot dt = \underline{\hspace{2cm}}$

d) $F'(x) = D_x \left[\int_1^x \frac{1}{t^2} \cdot dt \right] = D_x \left[\frac{-1}{x} + 1 \right] = D_x \left[-x^{-1} + 1 \right] = \underline{\hspace{2cm}}$

e) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_1^x \frac{1}{t^2} \cdot dt \right] = \underline{\hspace{2cm}}$

6) Let $F(x) = \int_1^x \frac{1}{t} \cdot dt$, $x > 0$

a) $\int \frac{1}{t} \cdot dt = \int t^{-1} \cdot dt = \underline{\hspace{2cm}}$

b) $F(1) = \int_1^1 \frac{1}{t} \cdot dt = \underline{\hspace{2cm}}$ Hint: What is the area from $t = 1$ to $t = 1$?

c) $F(2) = \int_1^2 \frac{1}{t} \cdot dt = \underline{\hspace{2cm}}$ Hint: Use Area Under Curve Program; Note: $\int_1^2 \frac{1}{t} \cdot dt = \int_1^2 \frac{1}{z} \cdot dz = \int_1^2 \frac{1}{y} \cdot dy = \int_1^2 \frac{1}{x} \cdot dx$

d) $F(3) = \int_1^3 \frac{1}{t} \cdot dt = \underline{\hspace{2cm}}$ Hint: Use Area Under Curve Program; Note: $\int_1^3 \frac{1}{t} \cdot dt = \int_1^3 \frac{1}{z} \cdot dz = \int_1^3 \frac{1}{y} \cdot dy = \int_1^3 \frac{1}{x} \cdot dx$

e) $F(4) = \int_1^4 \frac{1}{t} \cdot dt = \underline{\hspace{2cm}}$ Hint: Use Area Under Curve Program; Note: $\int_1^4 \frac{1}{t} \cdot dt = \int_1^4 \frac{1}{z} \cdot dz = \int_1^4 \frac{1}{y} \cdot dy = \int_1^4 \frac{1}{x} \cdot dx$

g) Use a scientific calculator and find $\ln(1) = \underline{\hspace{2cm}}$, $\ln(2) = \underline{\hspace{2cm}}$, $\ln(3) = \underline{\hspace{2cm}}$, and $\ln(4) = \underline{\hspace{2cm}}$

